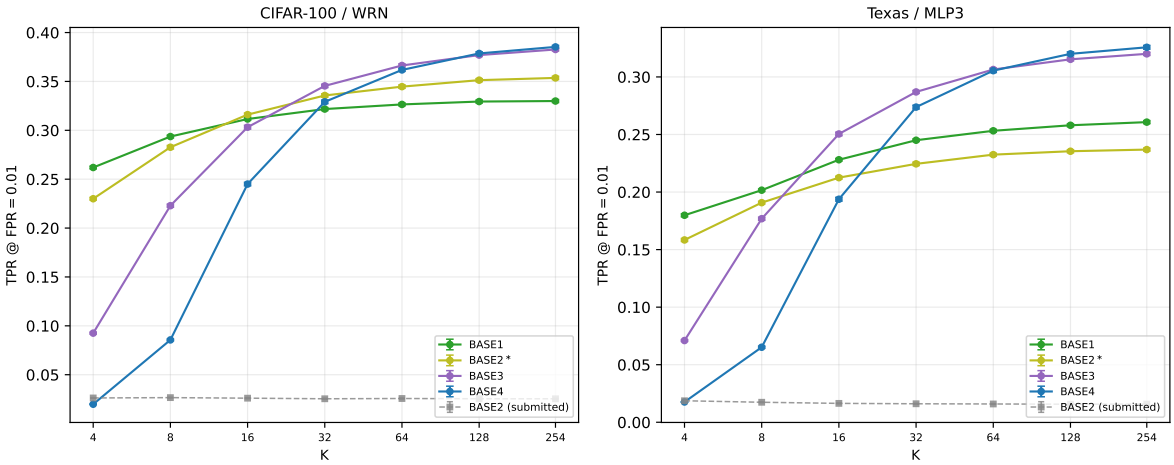

Figure 1: TPR @ FPR= 0.01 vs. shadow-model budget K (CIFAR-100 / WRN, Texas / MLP3). BASE1, BASE2*, BASE3, BASE4 (online) plus the submitted BASE2 (dashed grey). Submitted BASE2 sits near the floor across all K; BASE2* (separate IN/OUT means, constant variance) is the informative two-parameter intermediate. Mean ± 1 SE over 32 replicates. From `base32.pdf` (score formulas and full hierarchy there).

	$K = 8$ *(simple wins)*			$K = 254$ *(complex wins)*		
Testbed	AUC	TPR@.01	TPR@.001	AUC	TPR@.01	TPR@.001
CIFAR-10 / WRN	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**
CIFAR-100 / WRN	+0.71	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**
CINIC-10 / WRN	+0.08	+0.74	+0.79	**+1.00**	**+1.00**	**+0.94**
Location / MLP3	*-0.53*	+0.75	**+0.99**	**+1.00**	**+1.00**	**+1.00**
Purchase / MLP3	**+1.00**	**+1.00**	**+1.00**	+0.54	+0.83	+0.60
Texas / MLP3	**+1.00**	**+1.00**	+0.66	**+1.00**	+0.85	+0.73
CIFAR-10 / RN18	**+1.00**	**+1.00**	**+1.00**	*-0.54*	+0.71	+0.86
CIFAR-100 / RN18	-0.41	+0.38	+0.82	**+1.00**	**+1.00**	**+1.00**
CINIC-10 / RN18	-0.35	+0.55	+0.90	**+1.00**	**+1.00**	**+1.00**
Location / MLP4	-0.45	+0.37	+0.89	**+1.00**	**+0.99**	**+0.94**
Purchase / MLP4	**+1.00**	**+0.93**	+0.79	+0.90	**+0.90**	**+0.92**
Texas / MLP4	+0.57	**+1.00**	**+1.00**	**+1.00**	**+0.97**	+0.76
Mean	+0.39	+0.81	**+0.90**	+0.82	**+0.94**	+0.89
Median	+0.64	**+0.97**	**+0.94**	**+1.00**	**+1.00**	**+0.94**

Table 1: Weighted concordance $S \in [-1, +1]$ of the observed method ordering with the predicted BASE hierarchy. Reference at $K = 8$ is BASE1 > BASE2* > BASE3 > BASE4 (*simple wins*); at $K = 254$ the reverse (*complex wins*). $S = +1$ all pairs agree, -1 fully reversed. **Bold**: $S \geq 0.9$; *italic*: $S \leq -0.5$. Predicted ordering supported in $\geq 10/12$ testbeds for every tail metric.

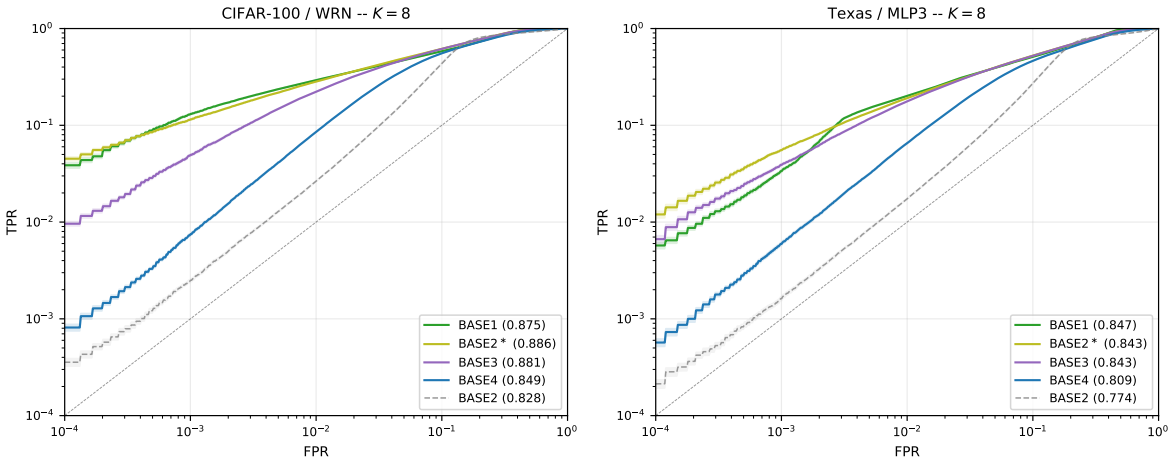

Figure 2: Online ROC at $K = 8$. At small K the simpler families lead in the deep tail (the *simple wins* regime; Table 1). From `base32.pdf`.

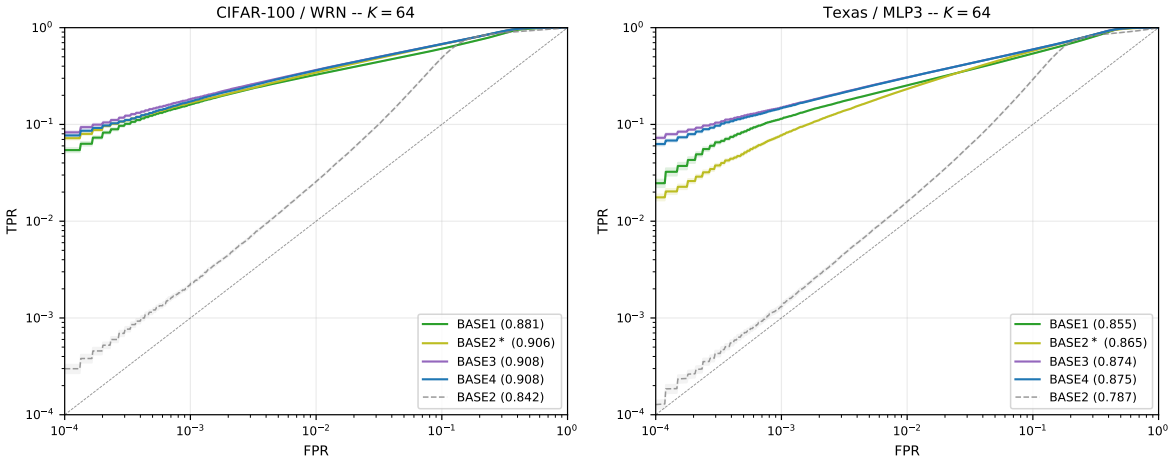

Figure 3: Online ROC at $K = 64$. The hierarchy is mixed at intermediate K. From `base32.pdf`.

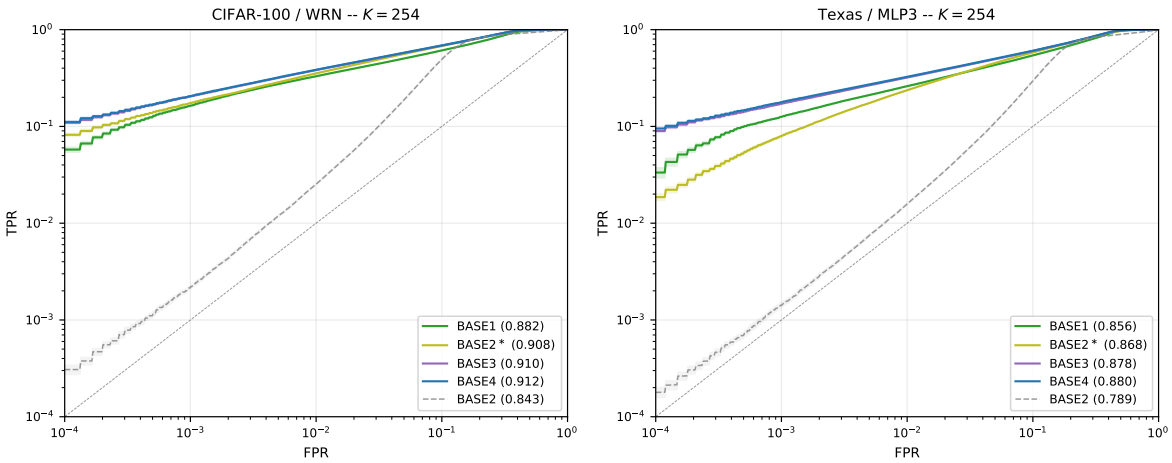

Figure 4: Online ROC at $K = 254$. At large K the richer families lead, reversing the small-K ordering (*complex wins*; Table 1). From `base32.pdf`.

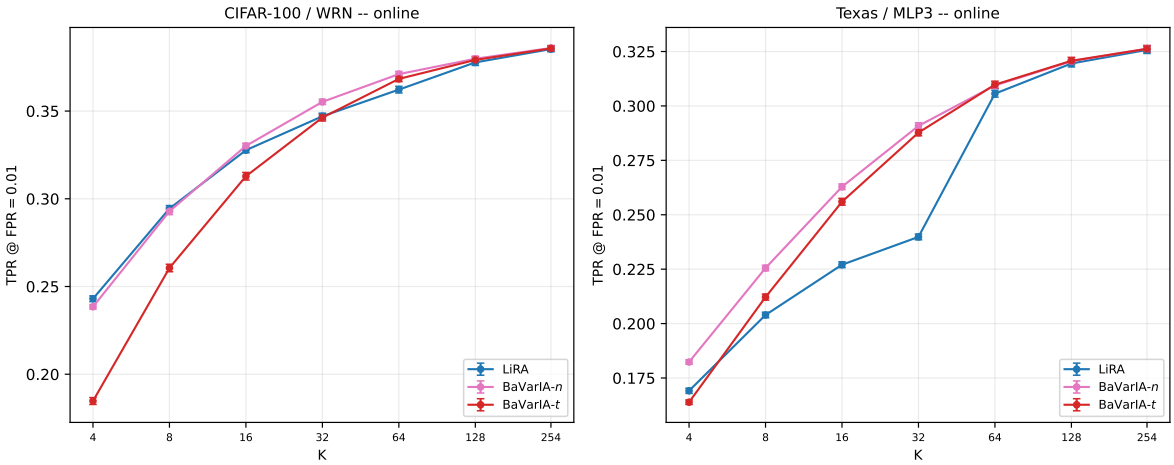

Figure 5: Online TPR @ FPR= 0.01 vs. K (CIFAR-100 / WRN, Texas / MLP3). LiRA, BaVarIA-n, BaVarIA-t. BaVarIA is a LiRA plug-in (NIG-shrinkage variance, continuous in K); the gain peaks around the $K = 64$ variance-rule cutover. Mean ± 1 SE over 32 replicates. From `bavaria32.pdf`.

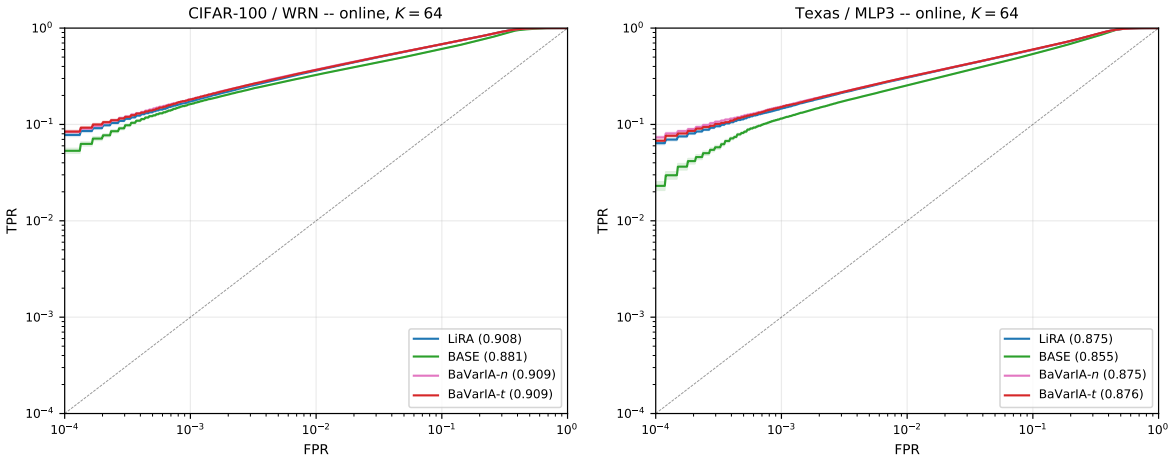

Figure 6: Online ROC at $K = 64$ (CIFAR-100 / WRN, Texas / MLP3). LiRA, BASE, BaVarIA-n, BaVarIA-t. Per-testbed breakdown in Table 2. From `bavaria32.pdf`.

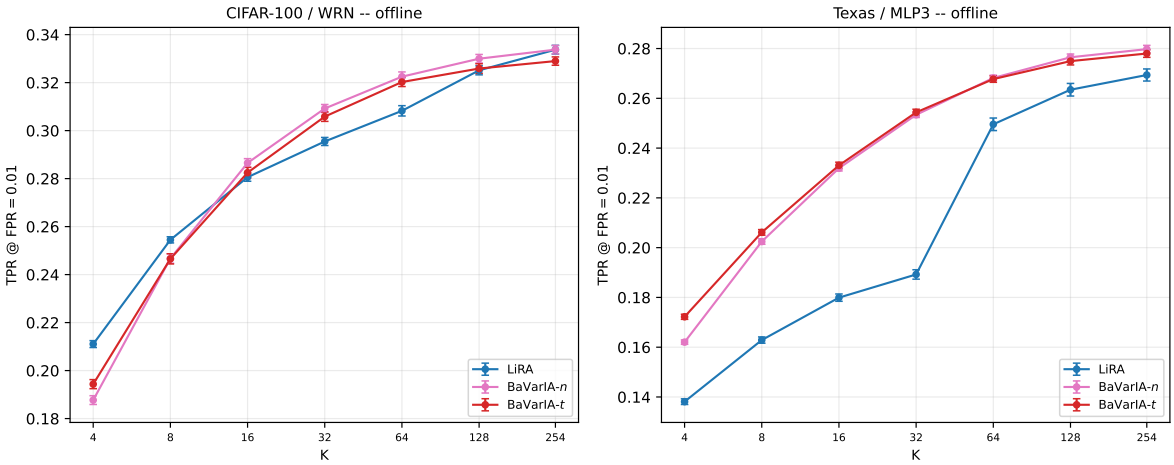

Figure 7: Offline TPR @ FPR= 0.01 vs. K (CIFAR-100 / WRN, Texas / MLP3). LiRA, BaVarIA-n, BaVarIA-t. Offline gap persists at every K (NIG prior substitutes for the IN class). Mean ± 1 SE over 32 replicates. From `bavaria32.pdf`.

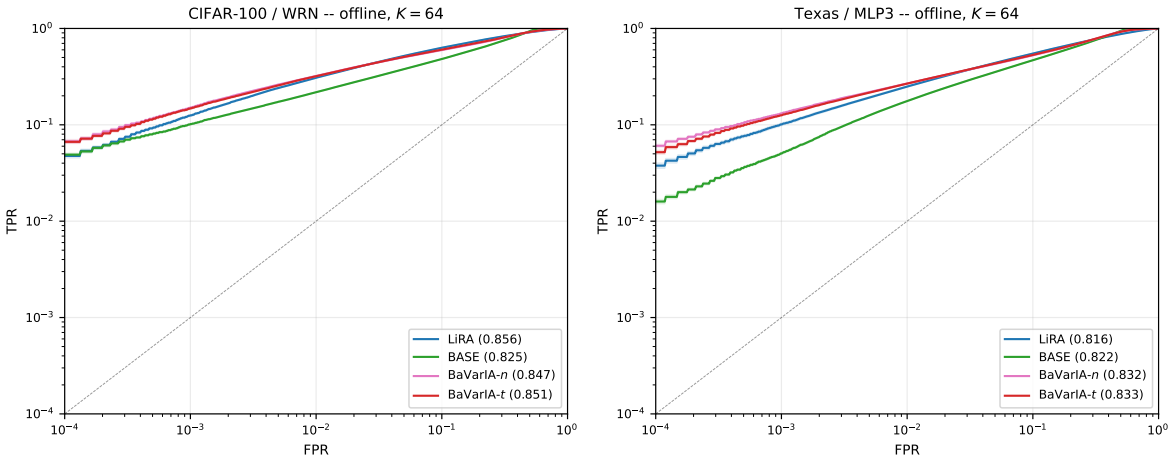

Figure 8: Offline ROC at $K = 64$ (CIFAR-100 / WRN, Texas / MLP3). LiRA, BASE, BaVarIA-n, BaVarIA-t. Per-testbed breakdown in Table 3. From `bavaria32.pdf`.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.129	**0.134**	0.133
CIFAR-100 / WRN	0.362	**0.371**	0.368
CINIC-10 / WRN	0.180	**0.187**	0.187
Location / MLP3	0.423	**0.440**	0.439
Purchase / MLP3	0.031	0.035	**0.036**
Texas / MLP3	0.306	0.309	**0.310**
CIFAR-10 / RN18	0.039	**0.042**	**0.042**
CIFAR-100 / RN18	0.663	0.669	**0.669**
CINIC-10 / RN18	0.288	**0.293**	0.291
Location / MLP4	0.352	**0.364**	0.361
Purchase / MLP4	0.053	0.059	**0.059**
Texas / MLP4	0.358	**0.359**	0.359

Table 2: Online TPR @ FPR= 0.01, $K = 64$ (LiRA's pooled-vs-per-sample variance-rule cutover). Per-testbed mean over 32 leave-one-out replicates, antithetic-paired design; best per row in bold. BaVarIA-n strictly beats LiRA on 12/12 testbeds, isolating the contribution of NIG-shrinkage variance interpolation. BaVarIA-t ties or trails BaVarIA-n at this K. Average of column means: LiRA 0.266, BaVarIA-n 0.272, BaVarIA-t 0.271.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.087**	0.074	0.069
CIFAR-100 / WRN	0.308	**0.322**	0.320
CINIC-10 / WRN	0.123	**0.181**	0.178
Location / MLP3	0.327	**0.412**	0.405
Purchase / MLP3	0.023	**0.026**	0.023
Texas / MLP3	0.250	**0.268**	0.268
CIFAR-10 / RN18	**0.038**	0.032	0.031
CIFAR-100 / RN18	0.579	0.663	**0.664**
CINIC-10 / RN18	0.223	**0.277**	0.273
Location / MLP4	0.299	**0.333**	0.326
Purchase / MLP4	0.038	**0.042**	0.039
Texas / MLP4	0.294	**0.321**	0.316

Table 3: Offline TPR @ FPR= 0.01, $K = 64$. Offline replaces the IN-class likelihood: LiRA discards it (heuristic rescaling of the OUT-only score); BaVarIA substitutes the NIG conjugate prior, giving a principled offline attack at LiRA-equivalent per-point cost. Per-testbed mean over 32 replicates, antithetic-paired design; best per row in bold. BaVarIA-n beats LiRA on 10/12 testbeds (LiRA wins only CIFAR-10 / WRN and CIFAR-10 / RN18). Average of column means: LiRA 0.216, BaVarIA-n 0.246, BaVarIA-t 0.242.